4 **(a)** Figure 4.1 shows a pan of water on a hotplate.

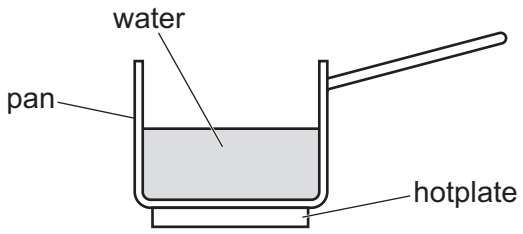

Figure 4.1

The mass of water in the pan is 200 g.
The specific heat capacity of water is 4.2 J/(g °C).

(i) The temperature of the water increases from 22 °C to 80 °C.

Show that the energy transferred to the water is approximately 49 kJ.

[2]

(ii) The time taken for the temperature of the water to increase from 22 °C to 80 °C is 41 s.

Calculate the minimum power of the hotplate.

power = ... W [2]

(iii) State **one** reason why the power of the hotplate must be greater than the value calculated in **4(a)(ii)**.

.....

..... [1]

DO NOT WRITE IN THIS MARGIN

(b) Describe, in terms of particles and their motion, **two** ways in which a gas differs from a liquid.

1 .....

.....

2 .....

.....

[2]

[Total: 7]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN